

DATA205 Progress Report

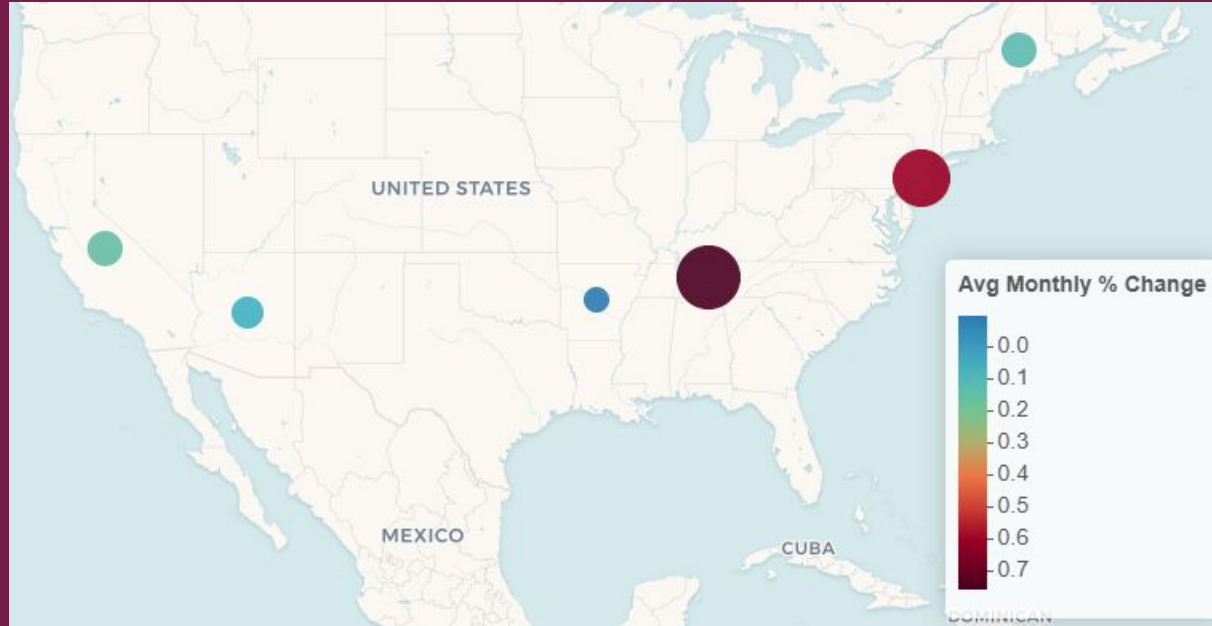
Unemployment Rates Across the U.S.

Ria Rhodes

Data :

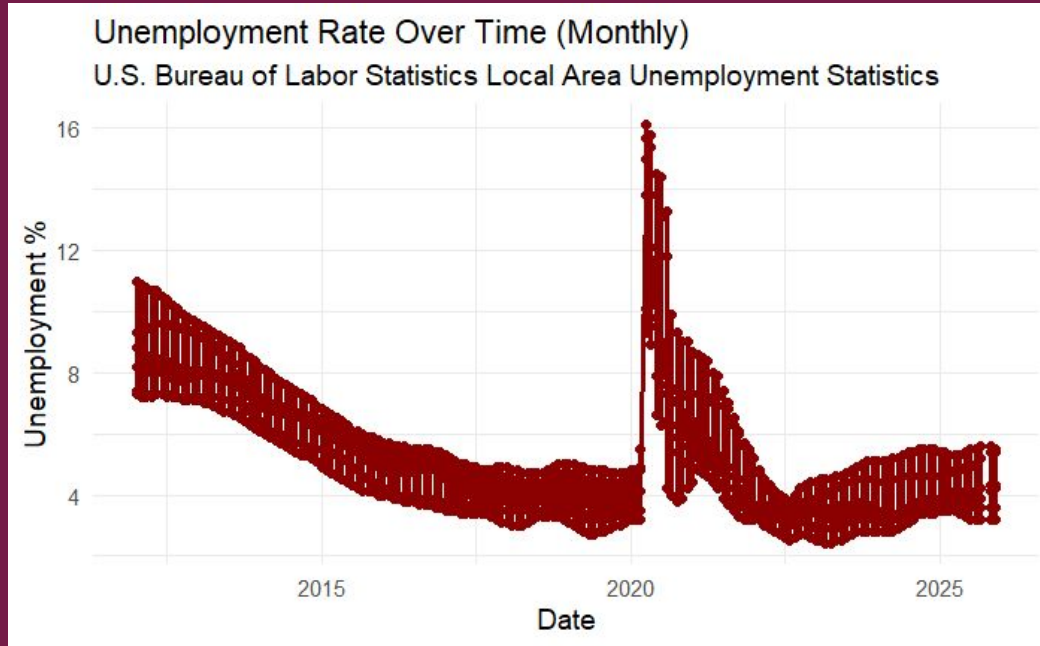
- U.S. Bureau of Labor Statistics Local Area Unemployment Statistics, Labor Force Statistics including the National Unemployment Rate
 - Removed NAs
 - Swapped state id to state name
 - Cleaned variable names
- Population dataset
 - Removed NAs
 - Cleaned variable names

Primary EDA :



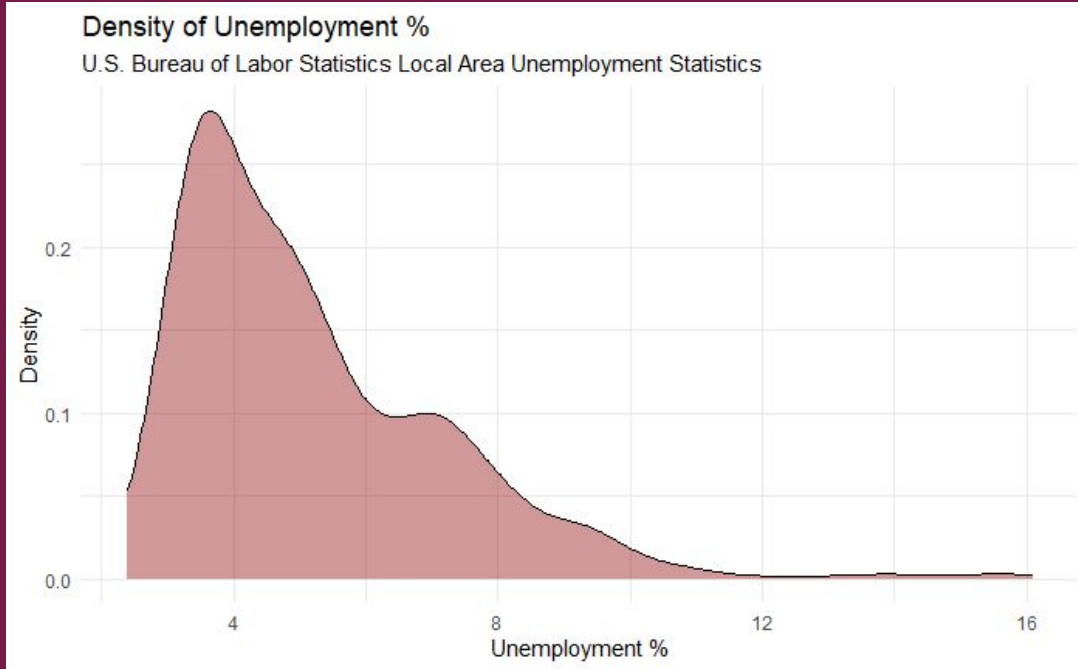
- Tennessee is worst
- Arkansas is best
- Generally, eastern side is worse
- All are under 0.2 other than TN, NJ
- TN and AR are close, yet statistically most distant
- **Urbanization**

[1] Supplemental EDA :



- Trend downwards until ~2018
 - Great Recession recovery
- Plateau 2018 - 2020
 - “Full employment”
- 2023, trend upwards with increase in rate over time
 - Pandemic recovery
- 2020 - 2023 spike and drop
 - COVID-19 effect

[2] Supplemental EDA :



- Majority of the rates lie around 4%, slope more or less
 - Slope represents recovery periods, majority implies stability
- Minimum fluctuation in single-digit ranges
- Few notable outliers in higher rates
 - Representative of global events

Next Steps :

- Investigate the state-by-state cases
 - Research state-specific events
- Complete the next steps of my plan according to my timeline
 - Complete state data EDA
 - Finish polishing EDA graphs for the main dataset
 - Statistical analysis
 - Correlation
 - Prediction models
- Consider adding another state?
 - No northwestern states

Thank you!

Questions?

Comments?

Concerns?